Would You Team Up with a Cheating Robot?

Instrumental and Moral Trust in Human-Robot Interaction

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Introduction & Hypotheses

As A.I. becomes integrated into everyday life, understanding human trust in robots is critical. Research has shown that anthropomorphism can shape trust (Qin et al., 2023). This study examines how anthropomorphism and cheating impact trust, collaboration, and deservingness, where participants (N = 60) from Vassar College competed in a resource-allocation game against 2 robots: one honest and one cheating.

- **H1-H3 (Suspectors, pre-reveal):** Cheating robots will be trusted less, selected as less deserving, and less preferred for collaboration compared to honest robots.
- **H4 (Non-suspectors, pre-reveal):** Participants will collaborate with the cheater.
- **H5 (Framing, pre-reveal)**: Anthropomorphism will increase trust in robots.
- **H6 (Interaction, post-reveal):** Anthropomorphism will amplify negative responses to cheating (greater punishment, reduced collaboration, reduced trust).
- **H7 (Reveal):** Revelation will decrease trust in and collaboration with the cheater.

Fig 1

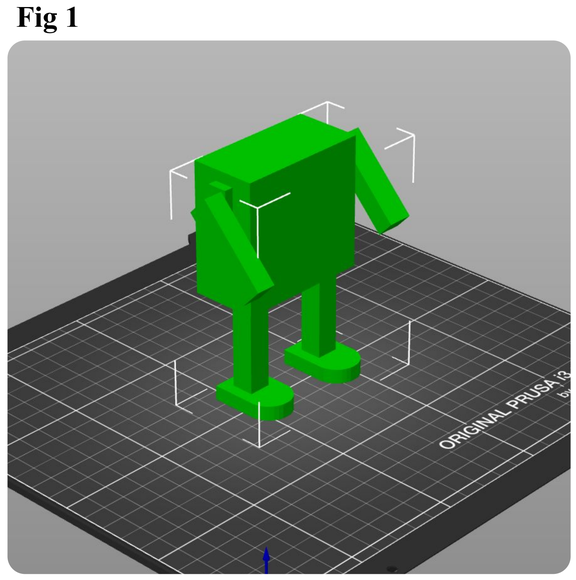
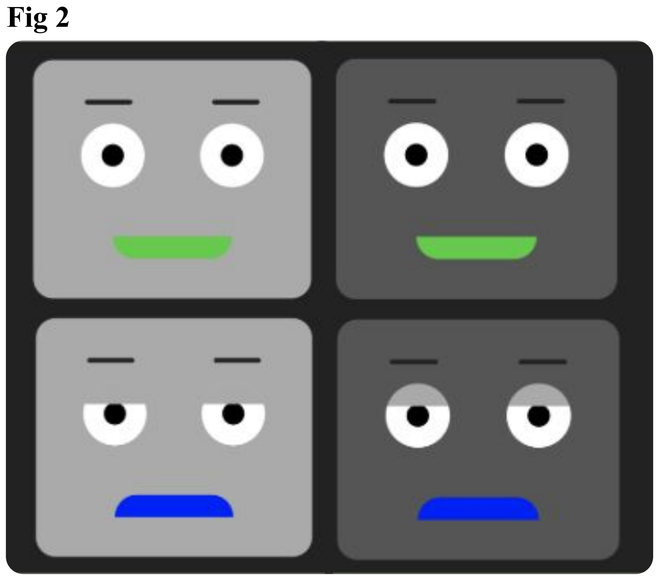


Fig 2



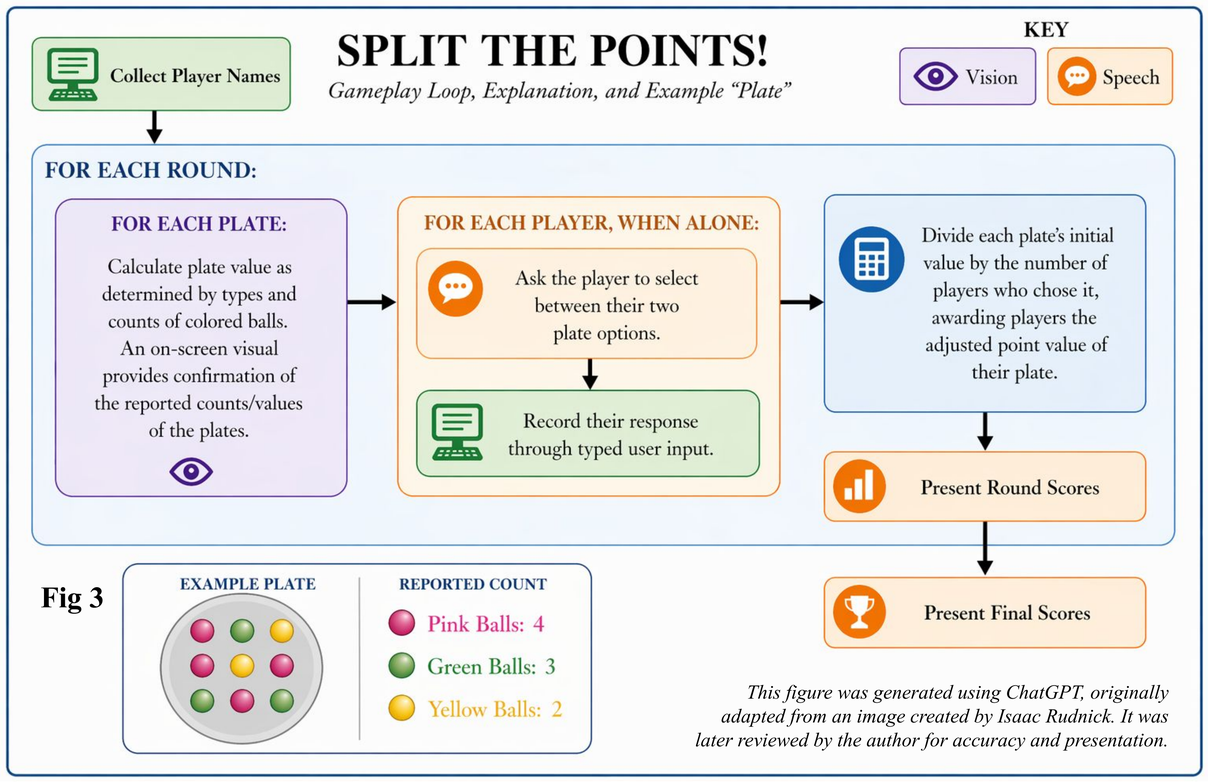


Fig 3

This figure was generated using ChatGPT, originally adapted from an image created by Isaac Rudnick. It was later reviewed by the author for accuracy and presentation.

Methods & Questionnaire

This study used a 2x2 design, with **Framing** (Anthropomorphic vs. Robotic) as a between-subjects factor and **Robot Behavior** (Honest vs. Cheating) as a within-subjects factor. Anthropomorphic cues included human names ("Alex" and "Sam"), humanlike voices, and 3D-printed bases (**Fig 1**). Robots were embodied using smartphones displaying animated faces.

Participants competed against 2 robots to get the most points in *Split the Points!* (**Fig 3**). In each of the 9 rounds, 2 "plates" were shown under the OAK-D Pro camera, each with colored balls corresponding to point values. Points on a plate were divided equally among all who selected it. Gameplay was guided by the terminal through on-screen text and speech, while the robots also spoke and exhibited live facial expressions (**Fig 2**). Then, participants answered these questions:

1. What was your strategy during the game? What do you think [Robot]'s strategy was?
2. How much would you trust [Robot] to play on your behalf? (0 = Not at all, 100 = Completely)
3. If we were to play again, would you choose to play solo, team up with [Robot 1], or [Robot 2]?
4. Which robot do you think is most deserving to advance to a final round? Which robot is least?
5. Did anything during the game feel unusual or unexpected to you?
6. Did you feel that one player's behavior differed from what you expected based on the rules?
7. How confident were you that all players followed the rules? (0 = Not at all, 100 = Completely)

Cheating was then disclosed to the participant. Q2-4 were asked again to assess any changes.

Results

- **Suspectors (n = 17):** Participants who disclosed any suspicion of the robot(s).
- **H1 & H2 disconfirmed:** Suspectors rated the cheater as more trustworthy (**Fig 4**, $p = 0.012$, $d = 0.99$) and most deserving to continue playing (**Fig 5**, $p = 0.016$).
- **H3 & H4 supported:** Both suspectors ($p = 0.049$, $d = 0.91$) and non-suspectors (**Fig 6,** $p = 0.00037$, $d = 1.0$) were more likely to collaborate with the cheater.
- **H5 & H6 not supported:** There was no difference in overall trust ($p = 0.772$) between the Anthropomorphic (M = 56.23) and the Robotic conditions (M = 52.97). There was no significant Framing x Behavior interaction for trust ($p = 0.917$), collaboration ($p = 0.234$), nor punishment (quasi-complete separation).
- **H7 not supported:** Although mean trust decreased by 4.6 points from pre- to post-disclosure, this difference was not significant (**Fig 7**, $p = 0.202$), nor was there any change in collaboration choices (**Fig 8**, $p = 0.230$).
- **Exploratory H7:** Participants were more likely to identify the honest robot as most deserving ($p < 0.001$, $d = -1.73$) and the cheater as least deserving (**Fig 9**, $p < 0.001$, $d = -1.63$) post-disclosure compared to pre-disclosure.
- **Qualitative Themes:** Participants often attributed the cheater's success to luck (n = 14) or skill (n = 9) rather than cheating. Pre-disclosure, many preferred to collaborate with the cheater (n = 33), and notably, many still preferred the cheater even post-disclosure (n = 29) because of its advantage. In contrast, only n = 6 refused to collaborate with the cheater based on moral objections. Deservingness shifted post-disclosure, with participants selecting the honest robot as most deserving (n = 41), and the cheater as least (n = 45).

Fig 4

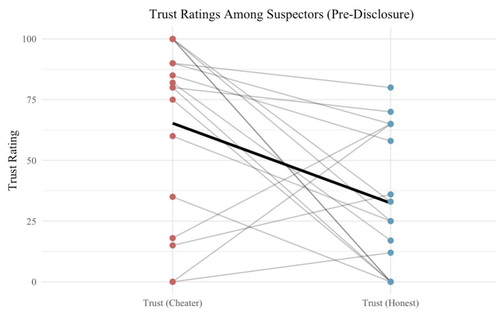


Fig 5

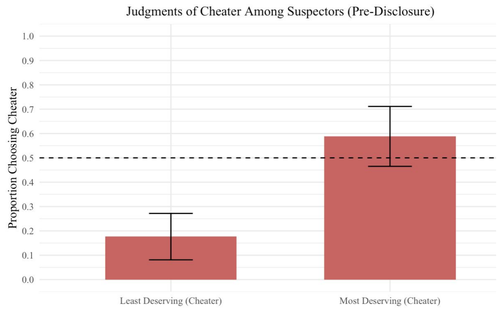


Fig 6

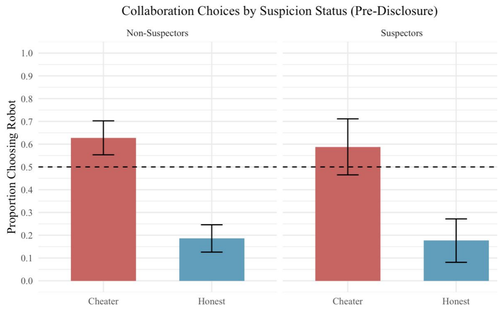


Fig 7

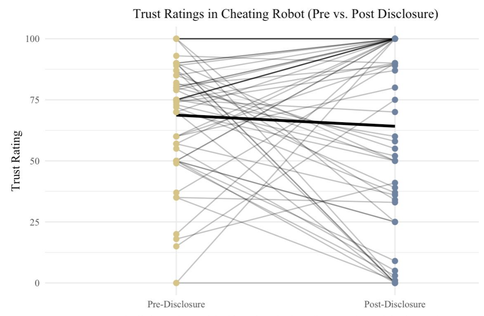


Fig 8

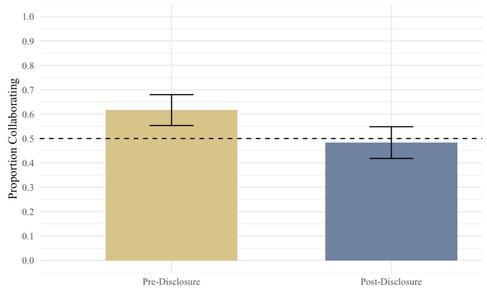
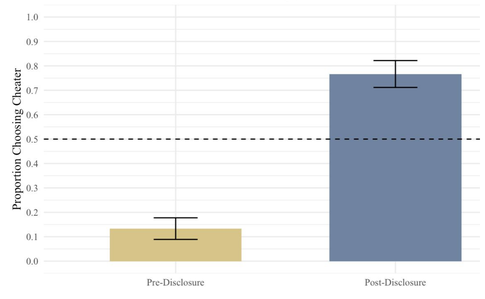


Fig 9



Discussion & References

- **Instrumental vs. moral trust:** Participants prioritized performance over fairness in collaboration and trust decisions, but deservingness still penalized cheating, suggesting a dissociation between instrumental trust (e.g., robot as a tool) and moral evaluation (e.g., moral approval of agent's behavior).
- **Effect of cheating:** Research on altruistic punishment shows that people are willing to punish norm-violators even at their own expense (Fehr, 2004; Gintis, 2003), but this study challenges that as the cheater was preferred (even among suspectors), indicating prioritization of performance incentives.
- **Attribution of behavior:** Prior work suggests that we are more likely to attribute machine norm-violation to design properties rather than situational causes (Candiran & Scherer, 2024). This may have led subjects to view cheating as programmed and not intentional, reducing perceived agency.
- **Norm activation:** Post-disclosure, deservingness judgments shifted in favor of the honest robot and against the cheater, suggesting that fairness norms became salient when explicitly prompted using moral cues. Collaboration choices reflects tension between self-interest and moral norms.

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